Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Deep Learning, Time-Series, Pattern Detection | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Professional Summary

Machine Learning Engineer with 10+ years across industry and academia, including 6 years building production ML systems in industry. I build machine learning pipelines end to end: data extraction, data mining, feature development, deep learning model training, testing, deployment, and monitoring of deployed models against the metrics that matter to the product. My work spans time-series and sequence modelling, computer vision at manufacturing scale, and anomaly scoring on large structured datasets. Published results include real-time detection in 14 ms on a Raspberry Pi 4 at F1 0.76, 74× faster than the baseline (JAES 2026); industrial computer-vision systems cut inspection time by 99.5% and lifted throughput by up to 300%. I evaluate models with baselines, ablations, hypothesis tests, and evaluation metrics such as ROC-AUC and F1, at the scale of 100,000 scored trials.

Core Competencies

ML Engineering: production ML pipelines (data extraction, feature development, model training, testing, deployment, monitoring); deep learning with TensorFlow/Keras and PyTorch; sequence models (RNN, LSTM, GRU) and embedding-based models; model optimization for latency and scalability
Domain Expertise: time-series and sequence data; content understanding from audio and signals (feature extraction, music information retrieval); computer vision; anomaly detection and scoring services; recommendation systems and ranking (working knowledge)
Research & Leadership: statistical evaluation (hypothesis testing, ROC-AUC, EER, precision/recall/F1, effect sizes); benchmark and ablation design; deep dives into individual components and overall ML architecture; cross-functional work with product, engineering and stakeholders; mentoring, CI/CD and code review

Technical Stack

Languages: Python, C++17, SQL, C, JavaScript/TypeScript, MATLAB, Shell
ML / AI: TensorFlow, Keras, PyTorch, scikit-learn, HuggingFace (Wav2Vec2), ONNX; recommendation and ranking systems (working knowledge)
Signal & Audio Processing: DSP, STFT, MFCC, CQT, chroma, Librosa, SciPy, FFTW, music information retrieval
Data & Cloud: AWS (EC2, S3, ECS, MWAA, RDS), Snowflake, SQL, ETL pipelines, NumPy, Pandas
Dev & MLOps: Docker, Airflow, Jenkins, Git, pytest, CI/CD, REST APIs; Kubernetes, MLflow (working knowledge)

Education

Ph.D, Information Engineering and Computer Science – *University of Trento, Italy* — *Jan 2025*
M.Sc, Telecommunication and Electronic Engineering – *Sheffield Hallam University, UK* — *Aug 2018*
B.Eng (Hons), Electronic Engineering – *Sheffield Hallam University, UK* — *Mar 2014*

Professional Experience

Independent Research & Open-Source | *Self-directed, Toronto, Canada* — *Jan 2026 - Present*

- Benchmarked 6 scoring methods across deterministic, probabilistic and deep-learning families (DTW, Gaussian likelihood, Wav2Vec2 embeddings) on 10,000 sentences and 100,000 scored trials; best model reached ROC-AUC 0.822 and EER 0.267 vs ROC-AUC 0.599 for the baseline.

- Built melodiq, a Dockerized audio-feature ETL pipeline: Airflow DAGs for data extraction and feature extraction, S3-to-Snowflake loading, infrastructure as code; built with AI-assisted workflows including Claude Code.

Postdoctoral Researcher | *University of Trento, Trento, Italy* *Jan 2025 - Dec 2025*

- Owned the end-to-end ML pipeline for streaming audio and IMU/gesture data on the EU Horizon EIC Pathfinder project MUSMET: data acquisition, feature engineering, model training and evaluation, deployment, and monitoring.
- Shipped real-time deep learning inference at 14 ms on Raspberry Pi 4, F1 0.76, 74× faster than the DTW baseline at 31.4% CPU (JAES 2026).
- Designed baseline and ablation suites (RNN vs DTW, audio vs MIDI, pattern-set sizes 1/3/10) that quantified accuracy, latency and CPU trade-offs and drove architecture decisions.
- Ran a live user study (20 audience members, 6 performers) with significance testing: lighting and haptics driven by real-time detection were rated significantly more coherent than random mappings ($p < .05$, IEEE I3DA 2025).

Visiting Researcher | *McGill University (IDMIL / CIRMMT), Montreal, Canada* *May 2024 - Aug 2024*

- Built multimodal IMU + audio recognition fused by a hierarchical state machine, P 0.79 / R 0.76 / F1 0.78 on a 500-event corpus (IEEE IS² 2025).

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka* *Aug 2020 - Feb 2021*

- Designed and deployed ML and statistical pipelines for financial forensics and risk detection on large structured enterprise datasets.
- Built SQL + Python backend scoring services for anomaly flagging and triage in production; translated compliance-stakeholder requirements into technical specifications.

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka* *Jan 2015 - Jul 2020*

- Built and shipped end-to-end ML and computer-vision systems for manufacturing, integrating camera, sensor and IoT streams into production workflows; operational efficiency up to +300%.
- Ran real-time ingestion and ETL for high-frequency IoT and operational time-series across sites in South Asia, North America and Africa, with C++/Python inference engines on embedded hardware.
- Delivered automated inspection (OpenCV, PLC-integrated reject) that cut inspection time 99.5%; designed the geometry-alignment algorithm and built the RIP integration for Promptly, now a Twinery/MAS on-demand printing product in four countries. Introduced CI/CD and code review; mentored engineers.

Selected Publications (full list: *nishal.xyz/publications*)

- Silva, N. and Turchet, L. *Real-Time Audio Pattern Detection for Smart Musical Instruments*. Journal of the Audio Engineering Society, Vol. 74, No. 3, Mar. 2026.
- Silva, N. and Turchet, L. *Music Information Retrieval as a Service: Latency Characterization of an Edge-to-Server Architecture*. IEEE International Symposium on the Internet of Sounds, 2026 (accepted).
- Silva, N. and Turchet, L. *Real-Time Pattern Recognition of Symbolic Monophonic Music*. 19th International Audio Mostly Conference (ACM), Sep. 2024.

Highlights & Recognition

- **Open datasets:** Four open-access datasets (9,800+ recordings, 70+ musicians) published on Zenodo as benchmarks for real-time pattern detection.
- **MIDI Innovation Awards:** Finalist (Sep 2023), Prototypes & Non-Commercial Software category, "Hot Licks" real-time pattern detection.
- **GMOA Recognition (Sri Lanka):** Computer-vision system for COVID-19 patient vitals monitoring deployed across hospital wards.